

Integrated Security Implementation Report

Module: ETTCS801 Cryptography and Network Security

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1 Introduction and Risk Assessment

A comprehensive baseline security audit revealed critical vulnerabilities at the ULK Polytechnic server architecture. This section outlines our systematic analysis and mitigation schema.

1.1 Risk Matrix and Controls

Table 1: Asset, Vulnerability, and Control Realignment

Asset	Vulnerability	Risk Level	Recommended Control
Student Records DB	Guest network accessibility	Critical (Rank 1)	VLAN isolation & firewall drops
Data in Transit	Unencrypted pipeline links	High (Rank 3)	Deploy TLS/SFTP protocol layers
Central Server Admin	Outdated software / Weak credentials	High (Rank 2)	Policy configuration, MFA & patching

2 Cryptography Application Engine

A dedicated tool engine was built using Python’s core processing primitives. This ensures symmetric file protection mapped with data verification checks.

2.1 Integrity Verification Logic

Data state is strictly checked utilizing the SHA-256 standard cryptographic hashing primitives to detect unauthorized subsequent modifications.

Listing 1: SHA-256 Verification Extraction Block

```
def calculate_sha256(filename):
    sha256_hash = hashlib.sha256()
    with open(filename, "rb") as f:
        for byte_block in iter(lambda: f.read(4096), b''):
            sha256_hash.update(byte_block)
    return sha256_hash.hexdigest()
```

3 Network Traffic Control Optimization

Traffic isolation constraints are processed dynamically through kernel state-level rule configurations via the host Netfilter daemon subsystem.

3.1 Rule Matrices Configured

```
sudo iptables -A INPUT -s 192.168.20.0/24 -d 192.168.1.100 -j DROP
sudo iptables -A INPUT -p tcp -s 192.168.10.0/24 --dport 5432 -j ACCEPT
sudo iptables -A INPUT -p tcp --dport 5432 -j DROP
```

4 Conclusion and Deliverables

The defenses applied isolate institutional records successfully from multi-vector compromise threats. Encryption guarantees data-at-rest protection, while strict firewall parameters close out architectural access leaks.

4.1 Repository Reference Location

GitHub Production Link: <https://github.com>